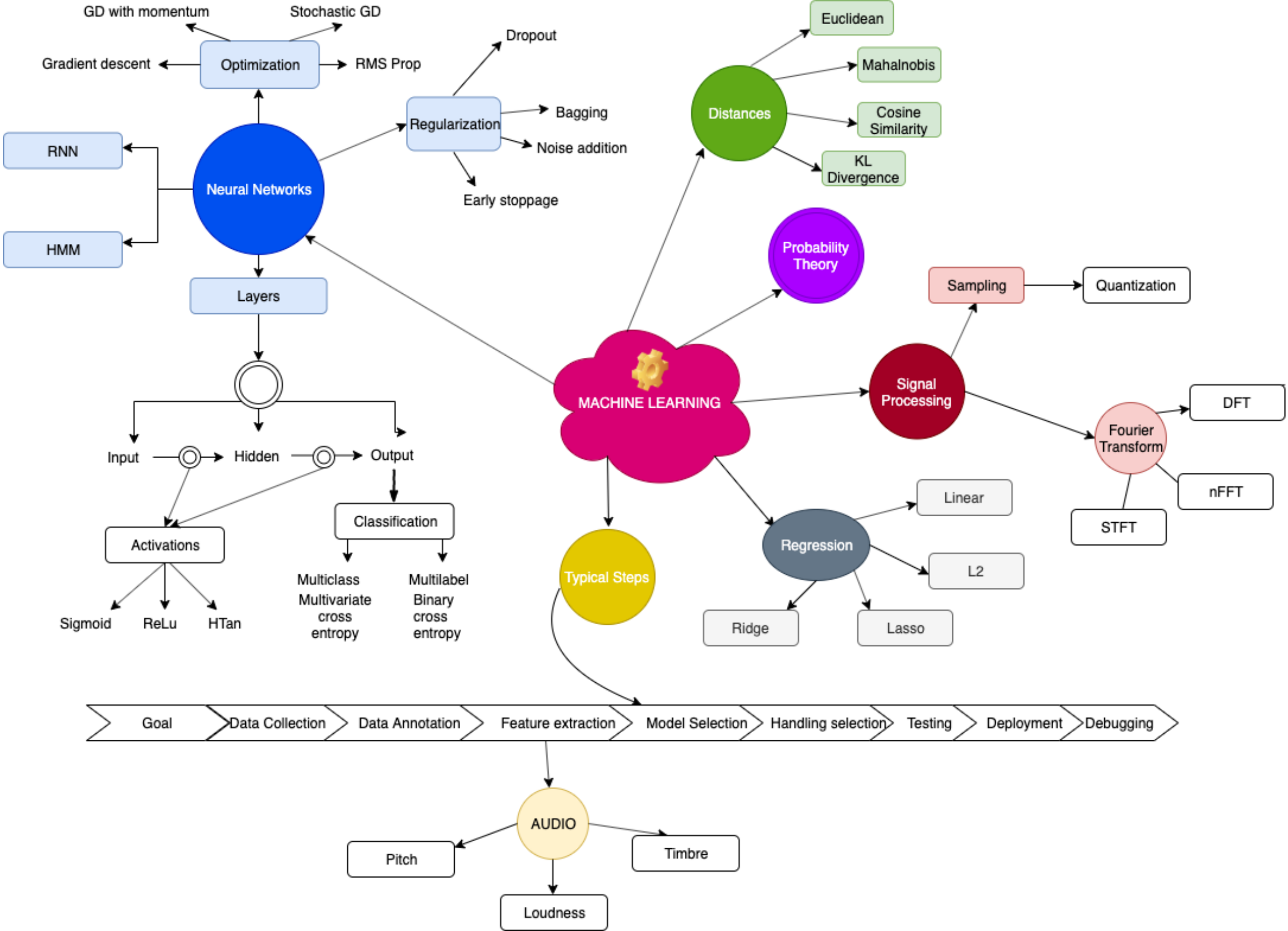
GD with momentum
Stochastic GD
Gradient descent
Optimization
RMS Prop
Dropout
Regularization
Bagging
Noise addition
Early stoppage
RNN
Neural Networks
HMM
Layers
Input
Hidden
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Activations
Sigmoid
ReLu
HTan
Classification
Multiclass
Multivariate
cross
entropy
Multilabel
Binary
cross
entropy
Euclidean
Mahalnobis
Distances
Cosine
Similarity
KL
Divergence
Probability
Theory
Sampling
Quantization
MACHINE LEARNING
Signal
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DFT
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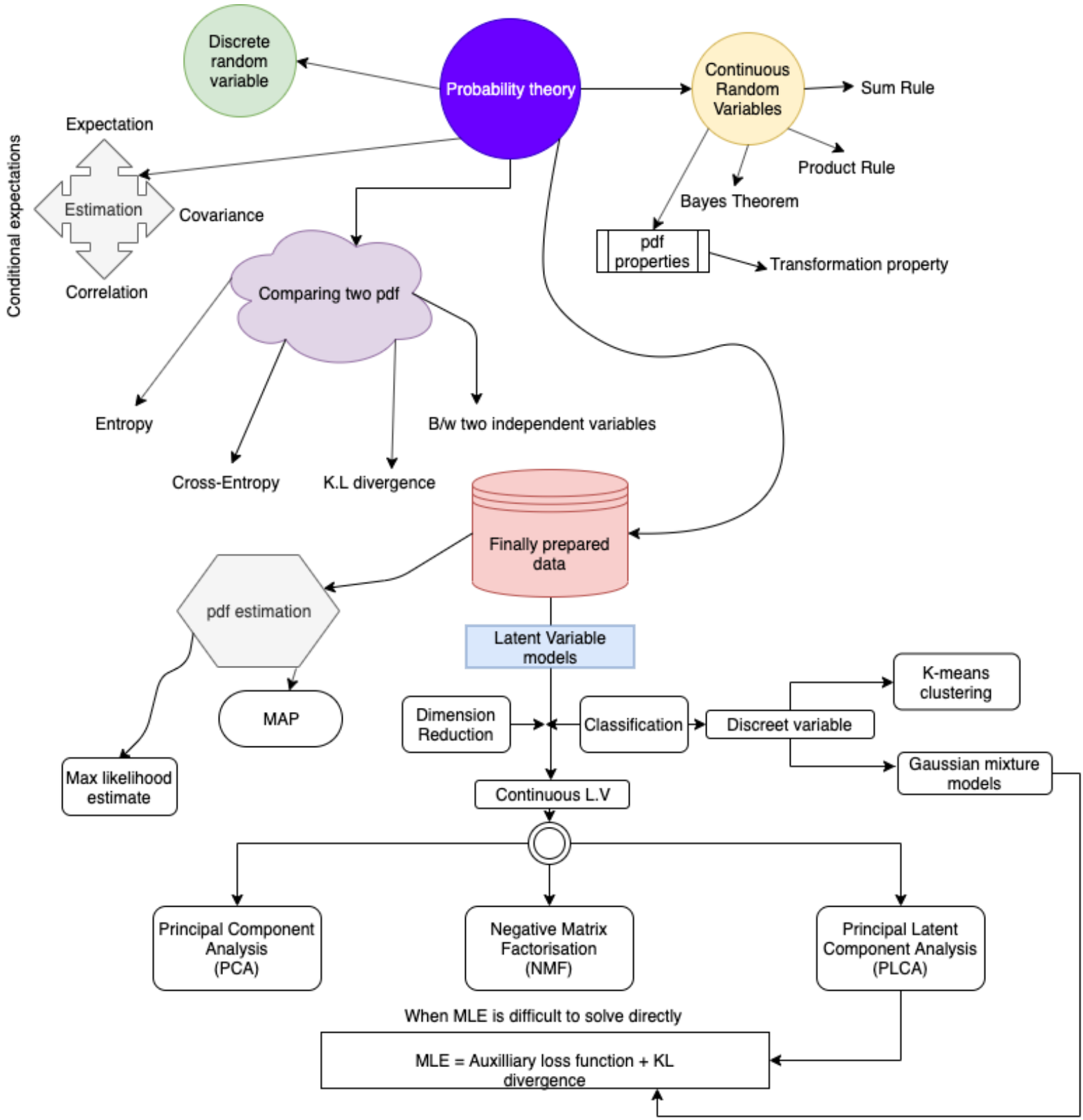
STFT
Linear
Regression
L2
Ridge
Lasso
Typical Steps
Goal
Data Collection
Data Annotation
Feature extraction
Model Selection
Handling selection
Testing
Deployment
Debugging
AUDIO
Pitch
Timbre
Loudness


Discrete random variable
Probability theory
Continuous Random Variables
Sum Rule
Product Rule
Bayes Theorem
pdf properties
Transformation property
Expectation
Estimation
Covariance
Correlation
Conditional expectations
Comparing two pdf
Entropy
Cross-Entropy
K.L divergence
B/w two independent variables
Finally prepared data
pdf estimation
MAP
Max likelihood estimate
Latent Variable models
Dimension Reduction
Classification
Discreet variable
K-means clustering
Gaussian mixture models
Continuous L.V
Principal Component Analysis (PCA)
Negative Matrix Factorisation (NMF)
Principal Latent Component Analysis (PLCA)
When MLE is difficult to solve directly
MLE = Auxilliary loss function + KL divergence